

Anwar Ben Brahim

Computer Engineering Student | Artificial Intelligence & Machine Learning |
Cybersecurity & Network Security

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PROFILE

Computer Engineering student at ENIT with hands-on experience in **Artificial Intelligence, Machine Learning and Cybersecurity** through academic, research and engineering projects. Interested in **AI-driven security, intelligent systems, network security and emerging AI applications**.

EDUCATION

Computer Engineering, <i>École Nationale d'Ingénieurs de Tunis (ENIT) — Tunis, Tunisia</i>	09/2024 – Present
Master's Degree in Systems and Communications (SYSCOM), <i>École Nationale d'Ingénieurs de Tunis (ENIT) — Tunis, Tunisia</i>	Sep 2025 – Present
Preparatory Cycle for Engineering Studies – Mathematics and Physics, <i>Institut Préparatoire aux Études d'Ingénieurs de Nabeul (IPEIN) — Nabeul, Tunisia</i>	09/2022 – 06/2024

TECHNICAL SKILLS

AI & Data

Machine Learning • Deep Learning • Data Mining •
Classification • Anomaly Detection • Feature
Engineering/Extraction • Explainable AI/SHAP

Programming & Frameworks

Python • Java • C/C# • Scikit-learn • NFStream • SHAP
• Node.js/Express • Java EE • MySQL

Cybersecurity & Networks:

Network Security • Intrusion Detection • Network
Traffic Analysis • Threat Detection • Incident
Response • Access Control • Malware Analysis

Tools

Linux • Docker • GitHub • Wireshark • AVISPA/ HLPSSL •
Ganache

PROFESSIONAL EXPERIENCE

RFC – Réseaux, Formation, Conseil, AI-driven MDR platform • Designed an AI-driven MDR platform for network threat detection and response in an isolated VirtualBox environment. • Extracted network-flow features using NFStream and developed a Random Forest classifier for normal and multiple attack categories. • Integrated SHAP Explainable AI for interpretable predictions and mapped threats to MITRE ATT&CK/D3FEND . • Developed a real-time detection dashboard and semi-automated response mechanism using Flask, Socket.IO and iptables . • Integrated a local LLM using Ollama/Phi-3 Mini to support security analysis.	06/2026 – 07/2026 Tunis, Tunisia
YUCCAINFO, Development of a real-time application for botnet detection • Analysis of attacks targeting the RDP protocol . • Design and deployment of a Docker-based testing lab to simulate attack scenarios, including brute-force attacks, DoS, and SSH tunneling. • Analysis of system logs to identify and investigate traces of attacks.	07/2025 – 08/2025 Sousse, Tunisia

- Study of **security mechanisms**, including firewalls, VPNs, and access-control measures.

OPUS LAB, *Software Development Intern*

Contributed to two web projects, focusing on application communication flows and collaboration within development teams.

06/2025 – 07/2025

Tunis, Tunisia

RESEARCH & ACADEMIC PROJECTS

Formal Validation and Implementation of Inference-Based Access Control

10/2025 – 04/2026

Systems in e-Health, *Final-Year Project / PFA2*

- Investigated **cybersecurity, privacy and access control** for secure e-health systems.
- Applied **formal verification** using **HLSL and AVISPA/OFMC** to validate security properties and analyse attack scenarios.
- Developed a blockchain-based security prototype using **Ethereum, Solidity, Ganache, Node.js and Express.js**.
- Designed a hybrid security architecture combining **blockchain-based access control and immutable audit logs** with off-chain storage of sensitive clinical data.

Biomedical Data Mining – EEG Seizure Detection

- Developed an end-to-end **data mining and machine learning pipeline** transforming raw EEG signals into features for automatic seizure-period detection.
- Applied signal preprocessing, analysis, feature extraction and classification to biomedical data.
- Evaluated the model's ability to distinguish seizure-related patterns.

Flood Risk Spatial Analysis for Coastal Regions

- Developed a **hybrid modelling approach** combining physics-based flood simulation with a **Random Forest classifier**.
- Performed spatial and environmental analysis for coastal flood-risk assessment.
- Developed an interactive **web mapping platform** for visualising spatial predictions.

Malware Analysis – PFA1

- Conducted static, dynamic and hybrid analysis of a ransomware variant in a controlled **FLARE VM/REMnux** laboratory.
- Used VirusTotal, PESTudio and Wireshark to investigate malware behaviour and network activity.

RESEARCH INTERESTS

- AI for Cybersecurity
- Network & IoT Security
- Intrusion & Anomaly Detection
- Explainable AI
- LLMs for Security
- Intelligent Threat Detection

ASSOCIATIONS & LEADERSHIP

ENIT Junior Entreprise, Senior Member / Client Project Manager
Managed client-oriented projects, contributed to two client projects and participated in professional events including Forum ENIT Entreprise.

09/2025 – 08/2026

ENIT Junior Entreprise, Membre actif
Participated in various student projects, contributing to the development of skills in project management, teamwork, collaboration, and professionalism.

10/2024 – 08/2025

SecuriNets ENIT
Participated in cybersecurity workshops and Capture The Flag (CTF) competitions.

CERTIFICATIONS

- Opus Lab – web development
- CCNAv7 – Introduction to Networks

SOFT SKILLS

Leadership & Initiative, Teamwork, Analytical & Critical Thinking, Problem Solving, Independent Learning, Adaptability, Scientific Curiosity, Communication & Negotiation

LANGUES

- Arabic
- English
- French